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WHO I AM

I am a 23-year-old engineer from Seoul who spends his days getting surgical robots to learn from surgeons' hands. I am finishing a double major in mechanical engineering and computer science at Seoul National University, I am a researcher at Seoul National University Hospital, and since March 2026 I have been the co-founder and CTO of Rosota, a spin-off of the hospital's MEDISC Lab. This year our team recorded surgeons operating with ordinary hand-held laparoscopic instruments, trained a policy on nothing but those recordings, and watched it complete appendectomies in three of four live rabbits. I was the first author of that study and I designed most of what made it possible: the sensor on the instrument, the pipeline that aligns its channels, the dual-arm platform that executes the policy, and the simulation protocol that chose the policy's design.

I was born in Indiana, grew up mostly in Korea, and am a dual citizen of the United States and the Republic of Korea, so I can study and work in either country without a visa. I write this because my friend at Boston University kindly offered to introduce me to a few groups in Boston, and I would like you to know who is walking into your office.

HOW I GOT HERE

I did not arrive at robotics through a lab. In 2021 I taught myself machine learning from textbooks and papers, re-implementing CNNs and RNNs in NumPy before I trusted any framework, and tutored other students in vision and language models. In 2022, as a sophomore, I started a company with three friends. Over three years we shipped five products: a RAG-based knowledge base, a chatbot that speaks as its owner (which reached #3 Product of the Day on Product Hunt), a real-time Korean web-data API for LLM applications, a browser-based cloud desktop controlled by a vision agent, and an open-source coding agent for hardware projects. Between products I worked at FriendlyAI on low-level GPU code for image-generation models.

Then I spent eighteen months in the Republic of Korea Army maintaining MD 500 attack helicopters, and came back in August 2025 wanting to work on machines that act in the physical world.

I joined Prof. Dongjun Lee's INRoL at SNU that autumn, just as NVIDIA released Isaac GR00T. I built a bridge between Isaac Sim and GR00T for the Fourier GR1 humanoid, with tutorials for fine-tuning and roll-out, pick-and-place and nut-pouring tasks, and an evaluation pipeline, and I wrote down everything that broke along the way. That "RFM Wiki" became the hands-on material of a graduate course on robot foundation models taught by Prof. Yongseok Lee (now at DGIST), where I was the teaching assistant in the fall of 2025. For my senior thesis in the same lab I work with Somang Lee on tactile sensing for humanoid hands: a fast tactile module whose stress propagation matches FEM to within 4–17% in peak pressure at half a millisecond per frame, and now a differentiable forward model that identifies a real sensor's physical parameters from robot indentation data.

WHAT I DID THIS YEAR

In December 2025 I joined the MEDISC Lab of Prof. Hyoun-Joong Kong at Seoul National University Hospital to work on autonomous laparoscopic surgery with Prof. Kyu Eun Lee, a surgeon. The lab had an idea I found compelling: most minimally invasive surgery is still done by hand, and the surgeon's instrument motion, the exact signal imitation learning needs, is thrown away when the operation ends. Instead of collecting

demonstrations on a robot, as SRT-H and similar work do, we could record them from the surgeon's own instruments, in the spirit of UMI but in an operating room. I took that idea from a sketch to a device, and the lab spun the work out as Rosota, where I became CTO.

The device is SurgiLogger, a clip-on recorder for standard laparoscopic instruments. An IMU, a time-of-flight rangefinder and a Hall sensor recover the four degrees of freedom the trocar leaves an instrument, plus the jaw, with nothing added to the part that enters the patient. The hard part was not the sensors but time. Each channel reaches the log with its own delay, and the depth channel lagged attitude by a third of a second. I measured every channel against a Franka FR3 driving the instrument through swept sines, rewound each by its own latency, and found that this one step raised closed-loop lift success from 19% to 81% while making the offline validation error slightly worse. That inversion shaped everything after it. We stopped selecting policies by offline error and instead ran 31 candidate configurations in closed loop inside an Isaac Sim scene reconstructed from depth maps of real organs; a frozen self-supervised vision backbone cut trocar violations from 88% to 6% relative to the usual ResNet encoder, and the configuration that won in simulation ran on the real two-arm rig without a single violation.

In the live-animal study the policy, retrained on 849 demonstrations from four rabbits, completed the appendectomy with electrosurgery in three of four further animals under the surgeon's supervision, with a peak remote-center error below one millimetre. All four animals were healthy a week later. I am proud of the result, but I am more careful about what it does not show: the surgeon still selected the surgical phase, because our vision-based phase predictor collapsed under the deployment endoscope, and eight rabbits on one day is a small study. The paper and the demonstration data are public at rosota.ai.

Between June and September our team submitted six papers to ICRA 2027. I am a co-author on all of them, first author on the in-vivo study and co-first author on a localization method that ranges between instruments with ultra-wideband carrier phase through the sterile drape, resolving the integer ambiguity with the trocar's geometry rather than a second radio channel. In the same months Rosota closed a seed round led by FuturePlay and was selected for Korea's TIPS program, and I filed the company's first patent and design applications as lead inventor.

WHAT COMES NEXT

Rosota's second phase starts in human operating rooms. With Prof. Kyu Eun Lee as principal investigator, our IRB application is under review and the first human case is planned for 20 November 2026. On the research side I am adding strain-gauge force sensing to SurgiLogger so that policies can learn how hard a surgeon pulls on tissue, not only where the instrument goes, and I am working on retargeting four-degree-of-freedom hand-held demonstrations to the six-degree-of-freedom wristed instruments of da Vinci-class robots, which is where most of the installed base is.

The questions I most want to spend a Ph.D. on sit at the seam between these threads. How do we scale demonstration data from the millions of hand-held operations performed every year, and what must be true of that data for a policy to transfer across instruments, tasks and platforms? How should force and tactile signals enter a surgical policy, and how do we calibrate the simulators we use to screen it? And how do we build the evidence that lets a surgeon safely share control with a learned system? I expect to graduate in August 2027 and I am thinking seriously about where to do doctoral work. I would also be glad to collaborate sooner: Rosota and the hospital give me a working data platform, two robot arms through trocars, and surgeons who want to run experiments.

I would be grateful for the chance to tell you about this work in person and to hear what your group is building.

Dongho Yee, Seoul, October 2026